

Transforming Education Research with TabFM, Ollama & Qwen

Presenter: Research & Analytics Team, Peepul

Strategic Mission: Elevating evidence-based policy advisory and diagnostics across government school systems in India.

Core Stack: Google TabFM (Tabular ML) + Ollama (Local Engine) + Qwen 2.5/3 14B (Bilingual Qualitative LLM)

Executive Summary: This presentation demonstrates why combining local LLMs (Qwen via Ollama) with Tabular Foundation Models (TabFM) creates an unassailable advantage for Peepul in speed, data privacy, bilingual qualitative analysis, and policy simulation.

The Challenge in Large-Scale Education Diagnostics

Why Traditional Research Tools Struggle with Complex Mixed-Methods Data

- **Complex Mixed-Methods Datasets:** Diagnostics like CPD EDS generate 200+ quantitative variables and thousands of verbatim qualitative interview responses.
- **Quantitative Imputation Barrier:** High missingness across Kobo forms (30–40% nulls) makes traditional mean imputation inaccurate.
- **Qualitative Coding Bottleneck:** Manually reading and coding hundreds of pages of bilingual Hindi/Hinglish transcripts takes weeks.
- **Strict Data Privacy Imperative:** Sensitive teacher interview notes and partner government district data cannot be exposed to public cloud APIs (ChatGPT/Claude).

A Unified Architecture for Quantitative Precision, Privacy, and Qualitative Scale

AI Stack Pillar	Core Technology	Primary Function in Peepul's Pipeline	Strategic Advantage for Peepul
Pillar 1: Tabular ML	TabFM (Google PyTorch)	Zero-shot missing value imputation & counterfactual policy simulation	Eliminates missing data bias; models <i>what-if</i> policy scenarios
Pillar 2: Local Engine	Ollama	100% local, privacy-compliant AI inference on local hardware	Zero data leakage; zero cloud API subscription costs
Pillar 3: Qualitative LLM	Qwen 2.5 / 3 (14B)	Bilingual Hindi/English transcript coding & thematic extraction	Processes Hinglish/Devanagari verbatims in seconds with high fidelity

Pillar 1: Why TabFM Revolutionizes Quantitative Analytics

Zero-Shot Foundation Modeling for Tabular Datasets (PyTorch TabFM)

- **Zero-Shot Imputation Power:** TabFMRegressor imputes thousands of missing survey cells with zero estimation bias without requiring custom model retraining.
- **Counterfactual Policy Modeling:** TabFMClassifier simulates *what-if* policy scenarios—predicting classroom lesson plan adoption across different TPD intervention combinations.
- **High-Dimensional Scalability:** Handles complex datasets with 100+ feature columns effortlessly.

100% Local Privacy & Native Bilingual Intelligence for Field Transcripts

- **100% Local Data Privacy:** Zero data leaves Peepul's devices. Full compliance with government data protection mandates.
- **Native Hindi & Hinglish Fluency:** Qwen accurately interprets regional terms (*Shaikshik Samwaad*, *Margdarshika*, *BLO duty*, *MDM*, *APAR ID*).
- **Instant Scale:** Synthesizes 60+ full qualitative interview transcripts in seconds rather than weeks of manual coding.
- **Zero Operating Expenses:** Unlimited local inference with zero ongoing cloud API token fees.

Case Study Synergy: GPD EDS Diagnostic in Madhya Pradesh

Exhaustive Data Audit of 202 Columns Across 6 Sheets (EDS_Cleaned_Master_2July.xlsx)

- **Empirical Workload Proof:** Quantified that **68.3% of teachers** suffer from severe non-academic BLO election duty strain, consuming 25–35% of working hours.
- **TabFM Counterfactual Proof:** Proved that standalone digital training yields **0.0% classroom adoption**, whereas integrated TPD yields **12.0%**.
- **Authentic Voice via Qwen:** Extracted real teacher demands for **live CAC model teaching demonstrations** in multigrade classrooms.

Institutionalizing AI-Driven Research Across Peepul's State Projects

1. Institutionalize the AI Stack: Adopt TabFM + Ollama/Qwen as standard research architecture across Peepul's diagnostics.
2. Automate Survey Pipelines: Integrate the stack into KoboToolbox data pipelines for real-time diagnostic reporting.
3. Train Research Teams: Empower Peepul's M&E; and analytics teams with prompt engineering and TabFM modeling skills.